

SAI TARRUN PITTA

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing and delivering scalable backend systems, distributed cloud platforms, and AI-enabled applications. Skilled in building high-volume payment platforms, cloud infrastructure, RAG retrieval pipelines, and secure production systems, with a strong focus on reliability, performance optimization, and security-first engineering.

TECHNICAL SKILLS

- **Languages:** Python, Go, JavaScript, TypeScript, SQL, Java, C++
- **Frontend Development:** React, Next.js, Tailwind, HTML/CSS, Redux.
- **Backend Development:** Node.js, FastAPIs, Django, Express, REST APIs, gRPC
- **Databases:** PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch
- **Cloud & DevOps:** Amazon Web Services (AWS), Google Cloud (GCP), Docker, Kubernetes, Terraform, GitHub Actions.
- **Testing & Quality:** Unit Testing, Integration Testing, Code Reviews, Test-Driven Development, CI/CD.
- **AI/ML:** TensorFlow, scikit-learn, LangChain, Agentic AI, RAG pipelines, Agentic Workflows, Prompt engineering.
- **Core Competencies:** Data Structures, Object-Oriented Design, Microservices, Scalability, Reliability, Observability

PROFESSIONAL EXPERIENCE

Pacific Life <i>Software Engineer Intern</i>	May 2025 - Dec 2025 <i>Newport Beach, CA</i>
• Automated up to 98% of claims triage workflows by architecting a serverless ML/LLM service with AWS Lambda, Amazon API Gateway, TensorFlow, and LangChain multi-agent orchestration. • Reduced claims processing time by 75% and workflow failures by 30% by implementing human-in-the-loop validation checkpoints across ML/LLM triage decisions. • Accelerated policy lookup by 98% and boosted claims processor productivity by 50% by engineering a secure RAG pipeline with LLM embeddings, Pinecone semantic search, source-cited responses, and access-controlled retrieval.	
Accenture <i>Software Engineer</i>	Jul 2022 - Jun 2024 <i>Hyderabad, India</i>
• Scaled a gRPC transaction microservice to process 2M daily transactions with 20K transactions/sec peak throughput and 99.99% uptime by implementing Protocol Buffers, idempotency keys, and Redis distributed locking. • Led AWS cloud infrastructure optimization using Amazon EKS auto-scaling and Amazon ElastiCache connection pooling, reducing compute 50% and cloud spend 35%. • Optimized SQL queries across distributed transaction systems using composite indexes and query refactoring, reducing latency 60% and database CPU usage 73%. • Designed reliability architecture with Amazon CloudWatch, Prometheus, and Splunk monitoring pipelines with automated alerts, reducing MTTR 25x.	
Cognizant <i>Software Engineer</i>	May 2021 - Apr 2022 <i>Hyderabad, India</i>
• Engineered ML churn prediction system using scikit-learn and TensorFlow with React dashboards across 50K customers, achieving 12% churn reduction and 87% forecast accuracy. • Developed scalable REST APIs and real-time transaction data pipelines for payment analytics with NLP-based support risk scoring, increasing throughput 50% while maintaining sub-500ms latency. • Secured payment application authentication with OpenID Connect and JWT, strengthening identity and access control while reducing unauthorized access risk 35%. • Automated CI/CD workflows for payment services using Docker and GitHub Actions reducing release lead time 50%.	

EDUCATION

California State University, Fullerton <i>Master of Science, Computer Science</i>	Aug 2024 - May 2026 <i>Fullerton, CA</i>
• Achievements: Publication: Hardware Trojan Detection with ML and Power Side Channels, IEEE CNS 2025 • Coursework: Advanced Algorithms, System Design, Machine Learning, Artificial Intelligence, Distributed Systems	

PROJECTS

LLM-Powered Knowledge Retrieval Platform
• Developed a distributed full-stack knowledge retrieval platform with semantic search, vector indexing, and NLP-driven query matching, improving information retrieval accuracy by 25% and reducing user search time by 40%.
Event-Driven E-Commerce Modernization
• Architected a monolith-to-microservices migration using the Strangler pattern, REST APIs, and event-driven design, enabling zero-downtime canary deployments, service-level isolation, and scalable order, payment, and inventory workflows for high-traffic e-commerce operations.